IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART II: MEng, BEng and ACGI

FIELDS

Tuesday, 26 May 10:00 am

Time allowed: 1:30 hours

There are THREE questions on this paper.

Question One carries 40 marks. Question Two and Question Three carry 20 marks each.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : R.R.A. Syms
Second Marker(s) : S. Lucyszyn

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Electromagnetic Fields 2020 – Formula Sheet

- **Differential operators (Cartesian co-ordinates)**

$\nabla = \partial/\partial x\ \underline{i} + \partial/\partial y\ \underline{j} + \partial/\partial z\ \underline{k}$

$\nabla\phi = \partial\phi/\partial x\ \underline{i} + \partial\phi/\partial y\ \underline{j} + \partial\phi/\partial z\ \underline{k}$

$\nabla\ .\ \underline{\mathbf{F}} = \partial \mathbf{F}_x/\partial x + \partial \mathbf{F}_y/\partial y + \partial \mathbf{F}_z/\partial z$

$\nabla\ x\ \underline{\mathbf{F}} = \{\partial \mathbf{F}_z/\partial y - \partial \mathbf{F}_y/\partial z\}\ \underline{i} + \{\partial \mathbf{F}_x/\partial z - \partial \mathbf{F}_z/\partial x\}\ \underline{j} + \{\partial \mathbf{F}_y/\partial x - \partial \mathbf{F}_x/\partial y\}\ \underline{k}$

$\nabla^2\phi = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$

- **Identity**

$\nabla\ x\ \nabla\ x\ \underline{\mathbf{F}} = \nabla\ (\nabla\ .\ \underline{\mathbf{F}}) - \nabla^2\underline{\mathbf{F}}$

- **Integral theorems**

$\iint_A \underline{\mathbf{F}}\ .\ d\underline{a} = \iiint_V \nabla\ .\ \underline{\mathbf{F}}\ dv$ (Gauss' theorem)

$\int_L \underline{\mathbf{F}}\ .\ d\underline{L} = \iint_A (\nabla\ x\ \underline{\mathbf{F}})\ .\ d\underline{a}$ (Stokes' theorem)

- **Maxwell's equations – integral form**

$\iint_A \underline{\mathbf{D}}\ .\ d\underline{a} = \iiint_V \rho\ dv$ (Gauss' Law)

$\iint_A \underline{\mathbf{B}}\ .\ d\underline{a} = 0$ (Magnetic equivalent of Gauss' law)

$\int_L \underline{\mathbf{E}}\ .\ d\underline{L} = -\iint_A \partial\underline{\mathbf{B}}/\partial t\ .\ d\underline{a}$ (Faraday's law)

$\int_L \underline{\mathbf{H}}\ .\ d\underline{L} = \iint_A [\underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t]\ .\ d\underline{a}$ (Ampère's Law)

- **Maxwell's equations – differential form**

$\mathrm{div}(\underline{\mathbf{D}}) = \rho$

$\mathrm{div}(\underline{\mathbf{B}}) = 0$

$\mathrm{curl}(\underline{\mathbf{E}}) = -\partial\underline{\mathbf{B}}/\partial t$

$\mathrm{curl}(\underline{\mathbf{H}}) = \underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t$

- **Constitutive equations**

$\underline{\mathbf{J}} = \sigma\ \underline{\mathbf{E}}$

$\underline{\mathbf{D}} = \varepsilon\ \underline{\mathbf{E}}$

$\underline{\mathbf{B}} = \mu\ \underline{\mathbf{H}}$

- **Electromagnetic waves (pure dielectric media)**

Time dependent vector wave equation $\nabla^2\underline{\mathbf{E}} = \mu_0\varepsilon\ \partial^2\underline{\mathbf{E}}/\partial t^2$

Time independent vector wave equation $\nabla^2\underline{E} = -\omega^2\mu_0\varepsilon_0\varepsilon_r\ \underline{E}$

For z-going, x-polarized plane waves $d^2E_x/dz^2 + \omega^2\mu_0\varepsilon_0\varepsilon_r\ E_x = 0$

Where $\underline{E}$ is a time-independent vector field

Plane wave solutions have the form $E(z) = E_0\ \exp(-jk_0z)$

Where $k = \omega\sqrt{(\mu_0\varepsilon_0\varepsilon_r)}$ is the propagation constant

Impedance of free space $Z_0 = \sqrt{(\mu_0/\varepsilon_0)}$

Refractive index $n = \sqrt{\varepsilon_r}$

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- **Power**

Instantaneous power flow $\underline{\mathbf{S}} = \underline{\mathbf{E}} \times \underline{\mathbf{H}}$

Time-averaged power flow $\underline{S} = 1/2\ \mathrm{Re}(\underline{E} \times \underline{H}^*)$

- **Transmission line formulae**

Transmission line equations for line with per unit length inductance L_p and capacitance C_p

$dV/dz = -j\omega L_p I$

$dI/dz = -j\omega C_p V$

Phase velocity and characteristic impedance of lossless line with per unit length inductance L_p and capacitance C_p

$v_{ph} = 1/\sqrt{(L_p C_p)}$

$Z_0 = \sqrt{(L_p/C_p)}$

Reflection and transmission coefficients at junction between lines of impedance Z_1 and Z_2

$R_V = (Z_2 - Z_1)/(Z_2 + Z_1)$

$T_V = 2Z_2/(Z_2 + Z_1)$

Input impedance for length d of line with properties (Z_0, k) terminated by load Z_L

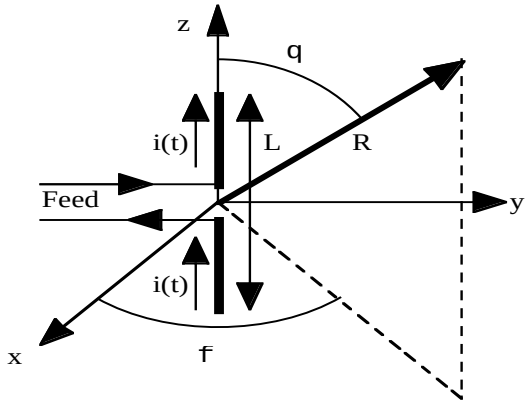
$Z_{in} = Z_0 \{Z_L + jZ_0 \tan(kd)\} / \{Z_0 + jZ_L \tan(kd)\}$

- **Antenna formulae**

Far-field pattern of half-wave dipole

$E\theta = j\ 60 I_0 \{\cos[(\pi/2)\cos(\theta)] / \sin(\theta)\} \exp(-jk_0 R)/R$; $H\phi = E\theta/Z_0$

Here I_0 is peak current, R is range and $k = 2\pi/\lambda$



Time averaged power flow $\underline{S} = 1/2\ \mathrm{Re}\ (\underline{E} \times \underline{H}^*) = S(R, \theta)\ \underline{r}$

Normalised radiation pattern $F(\theta, \phi) = S(R, \theta, \phi) / S_{max}$

Directivity $D = 1/ \{1/4\pi \iint_{4\pi} F(\theta, \phi) \sin(\theta)\, d\theta\, d\phi\}$

Gain $G = \eta D$ where η is antenna efficiency

Effective area $A_e = \lambda^2 D/4\pi$

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- **Transmission formulae**

Friis transmission formula $P_R = P_T \eta_T \eta_R A_T A_R/(R^2\lambda^2)$

Radar equation $P_R = P_T (\eta_T^2 A_T^2 \sigma/4\pi R^4\lambda^2)$

- **Microwave formulae**

Skin depth $\delta = 1/\sqrt{(\pi f \mu_0 \sigma)}$

- **Optical formulae**

Snell's law $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$

Fresnel reflection coefficient $\Gamma_E = \{n_1 \cos(\theta_1) - n_2 \cos(\theta_2)\}/\{n_1 \cos(\theta_1) + n_2 \cos(\theta_2)\}$

- **Imaging formulae**

Spherical wave $E(r) = (E_0/r) \exp(-jk_0 r)$ where $r^2 = x^2 + y^2 + z^2$ in Cartesian coordinates

Paraxial wave $E(R, z) = A(z) \exp(-jk_0 R^2/2z)$ where $R^2 = x^2 + y^2$ in Cartesian coordinates

The lens maker's formula $1/f = (n - 1) (1/R_1 + 1/R_2)$

The imaging equation $1/u + 1/v = 1/f$

- **Constants**

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m (permittivity of free space)

$\mu_0 = 4\pi \times ^{-7}$ H/m (permeability of free space)

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1. a) Identify the displacement current and normal current terms in Ampere's law. Estimate their ratio at angular frequency ω in a material of relative dielectric constant ε_r and conductivity σ. Assuming that $\varepsilon_r = 11.7$ and $\sigma = 1.56 \times 10^{-3}$ S/m for intrinsic Si, calculate this ratio at 1 MHz and 1 GHz.

[8]

b) Using Maxwell's equations, explain why there can be no electric or magnetic field outside a coaxial cable.

[8]

c) Sketch the cross-section of a microstrip waveguide and the variation of the electric and magnetic field lines. Derive simple expressions for the per-unit length capacitance and inductance. Use these to estimate the conductor width needed for 50 Ω characteristic impedance if the substrate has a relative dielectric constant of 4.4 and a thickness of 1.6 mm.

[8]

d) A dielectric slab of refractive index $n_1 = 1.5$ and thickness $d_1 = 100$ μm as shown in Figure 1.1a is to be coated with an additional layer of material of refractive index $n_2 = 1.33$ and thickness $d_2 = 10$ μm as shown in Figure 1.1b. Calculate the new slab thickness d_1' needed if the optical transfer function is to remain unaltered. Ignore loss and any reflections.

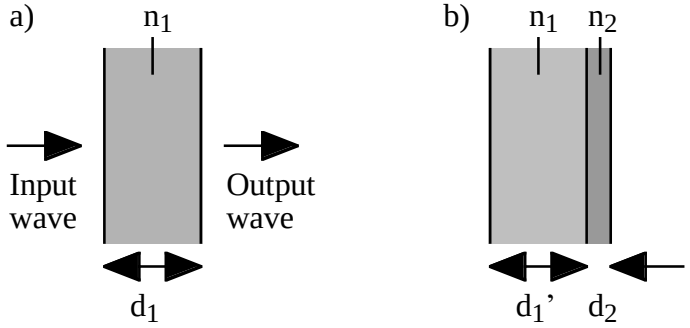


Figure 1.1.

[8]

e) The TE reflection coefficient at normal incidence from air onto a particular glass is -0.2. Find the refractive index of the glass, and the reflection coefficient at an angle of 30°.

[8]

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2. For a particular medium, the variation between the angular frequency ω and the propagation constant k for electromagnetic waves takes the form:

$$\omega^2 = \omega_P^2 + c^2k^2$$

Here, ω_P is the plasma frequency and c is the velocity of light in free space. Explaining your reasoning:

a) Sketch the dispersion characteristic, indicating the frequency ranges within which waves do and do not propagate.

[6]

b) Derive analytic expressions for the variations of the phase velocity v_{ph} and group velocity v_g with ω, and sketch these variations. To what values do v_{ph} and v_g tend when ω is large?

[8]

c) Derive an analytic expression for the propagation constant below cutoff, and explain its meaning. The plasma frequency is 1 GHz. Calculate the distance to which a wave will propagate before its amplitude falls to 1/e of its initial value, at a frequency of 0.999 GHz.

[6]

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3. a) The scalar wave equation for electric fields that only vary with radius r can be written as;

$$d^2E/dr^2 + (2/r)\, dE/dr + \omega^2\mu_0\varepsilon_0 E = 0$$

Show that the diverging spherical wave $E = (E_0/r) \exp(-jk_0r)$ is a solution, find the value of k_0, and explain how the solution satisfies power conservation.

[5]

b) Briefly explain the meaning of antenna directivity. Calculate the power density obtained at a distance of 10 km from a radio transmitter of 2 kW power, assuming i) an ideal, lossless isotropic antenna, and ii) a real antenna with an efficiency of 80% and a directivity of 100.

[5]

c) Assuming that $r^2 = z^2 + R^2$ and $R^2 = x^2 + y^2$, devise a paraxial approximation for the spherical wave solution, valid when $R^2 << z^2$. How does the phase of the wave vary with R? How should your solution be modified for a converging wave?

[5]

d) The transfer function of a lens with focal length f can be written as:

$$\tau_L = \tau_S \exp\{jk_0R^2/(2f)\}$$

where τ_S is the transfer function of a dielectric slab. Show that the lens transforms one paraxial wave into another, and prove the imaging equation $1/u + 1/v = 1/f$, where u and v are the object and image distances. Find the image distance and magnification when the object is 200 mm away from a lens with focal length 100 mm.

[5]

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